Technically, second-generation drones incorporated radio-controlled interfaces that allowed operators to manage altitude, direction, and speed over greater distances. The integration of GPS navigation enabled autonomous waypoint flight paths, automatic return-to-home functions, and improved spatial accuracy. Drones in this generation typically featured an onboard flight controller, inertial sensors, and communication modules that facilitated real-time telemetry and data feedback to the operator. The materials used for construction shifted toward lightweight composites and improved aerodynamics, allowing longer flight durations and higher payload capacities.

This generation also introduced early forms of camera payloads and sensor integration, making drones capable of performing tasks such as aerial mapping, reconnaissance, and environmental observation. The use of digi-tal communication links improved command reliability, while advances in power systems, particularly with lithium-based batteries, enhanced opera-tional efficiency.

Figure 1.4 illustrates the architecture and key components of a second-generation drone, highlighting the integration of GPS modules, radio control receivers, and flight controllers. The diagram depicts the interaction between the onboard systems and the ground control station, emphasizing how satellite-based navigation and electronic stabilization enabled more accurate and automated flight paths. This figure reflects the technological evolution that bridged the gap between manual drones and modern autonomous UAV platforms.